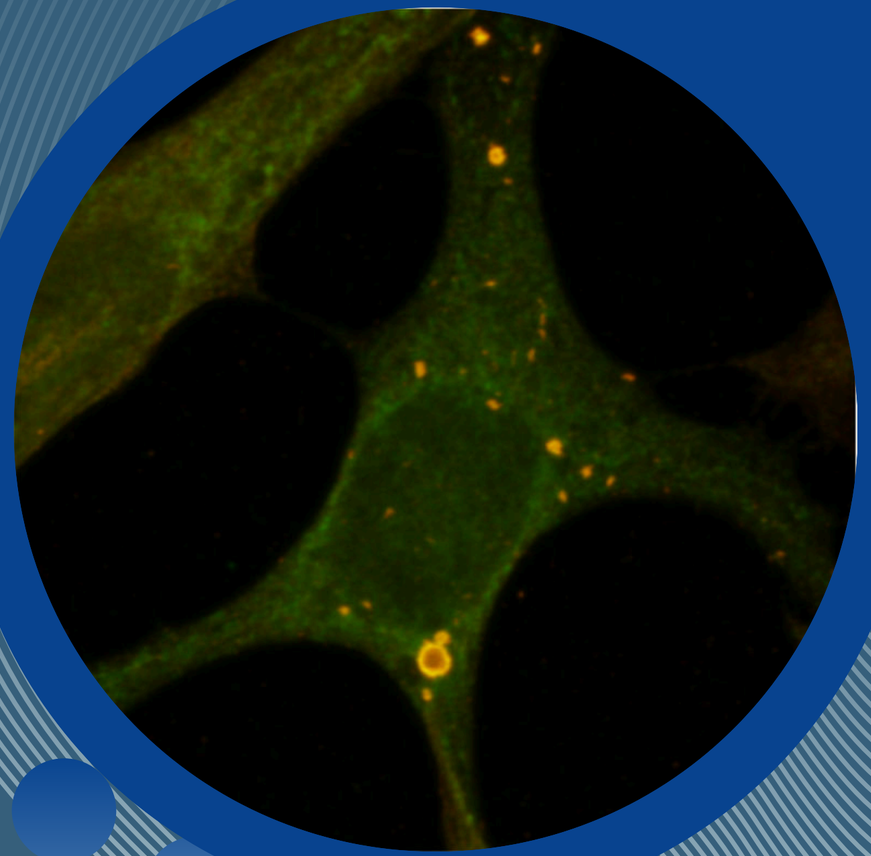
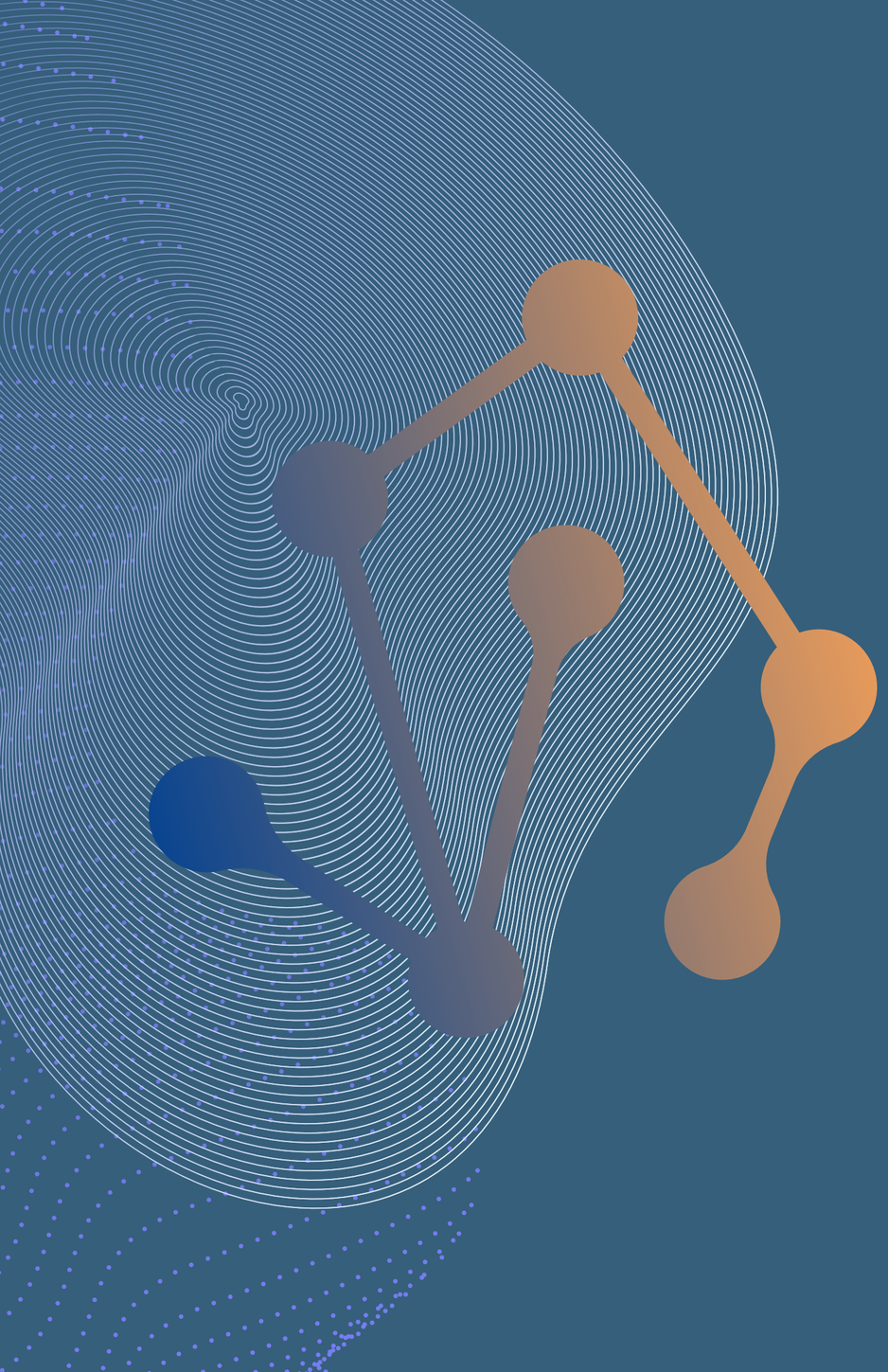
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Quantitative Colocalization Analysis of CCT1 and Mutant Huntingtin Aggregates



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Introduction

- **Noah Wijnheijmer**
- **Master Applied Artificial Intelligence**



PREDICTING, SLOWING DOWN, AND CURING HEREDITARY BRAIN DISEASES

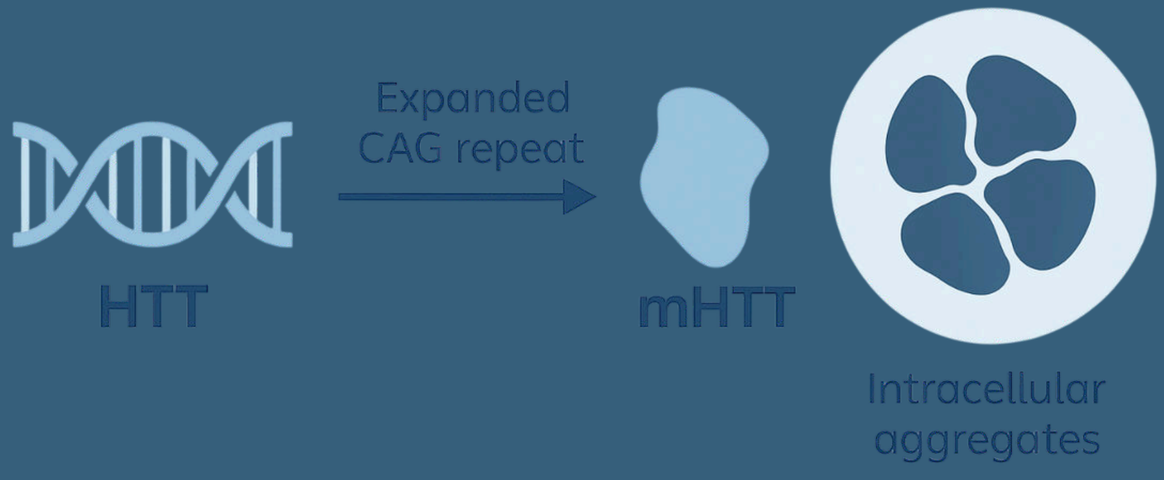
Through this research, I aim to unravel interactions between the **CCT1 chaperone** and **mutant Huntingtin**



Context

Background on Huntington's Disease (HD):

- Fatal, inherited neurodegenerative disease.
- Caused by CAG repeat expansion in the *HTT* gene.
- This produces a misfolded mutant huntingtin protein (mHTT), which forms intracellular aggregates.



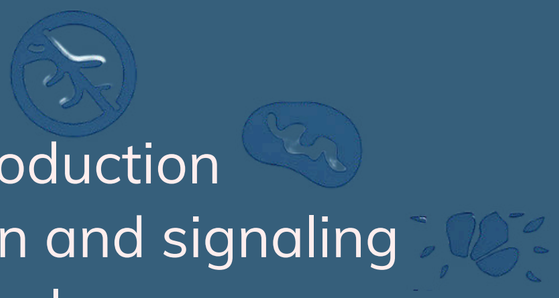
Context

Cellular Impact of mHTT Aggregates:

Aggregates disrupt:

- Transcription regulation
- Mitochondrial energy production
- Neuronal communication and signaling

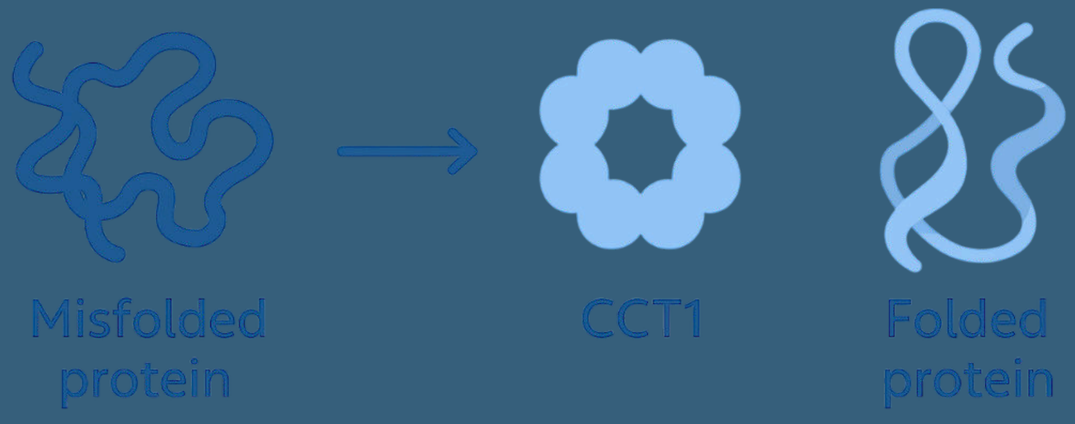
In both the nucleus and cytoplasm



Context

The Protective Role of CCT1:

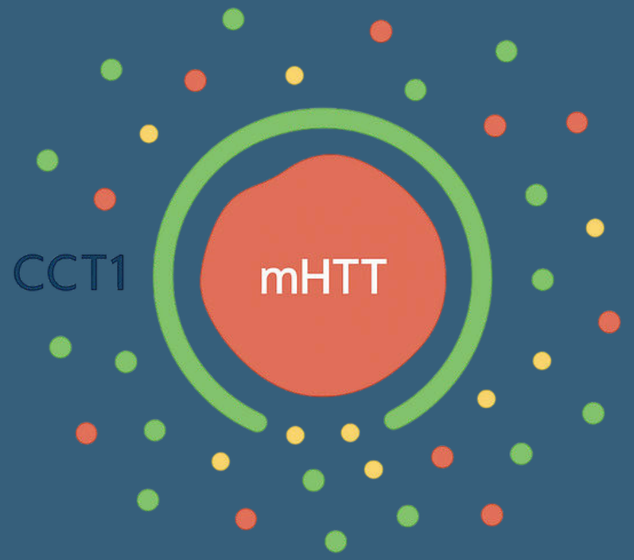
- Chaperones like CCT1 assist in proper protein folding and help prevent misfolding and aggregation.
- Elevated CCT1 levels have been shown to reduce mHTT aggregation and improve neuronal survival in cell models.



Context

TESTING CCT1 RECRUITMENT VIA COLOCALIZATION:

- WHERE DOES CCT1 ACCUMULATE AT MHTT AGGREGATION SITES?
- IF CCT1 IS RECRUITED TO MHTT AGGREGATES, SPATIAL COLOCALIZATION SHOULD BE OBSERVABLE.
- COLOCALIZATION PROVIDES INDIRECT EVIDENCE SUPPORTING A POSSIBLE ROLE OF CCT1 IN PROTEOSTASIS REGULATION.



Research question

“Does the molecular chaperone CCT1 show significant spatial enrichment at mutant Huntingtin (mHTT) aggregate sites, and can this relationship be robustly quantified using automated analysis?”

Problem statement

Quantifying CCT1's spatial relationship within huntingtin cells

Current methods are:

- Time consuming
- Labor intensive
- Subject to human bias
- Limited in reproducibility across experiments

Data and Setup

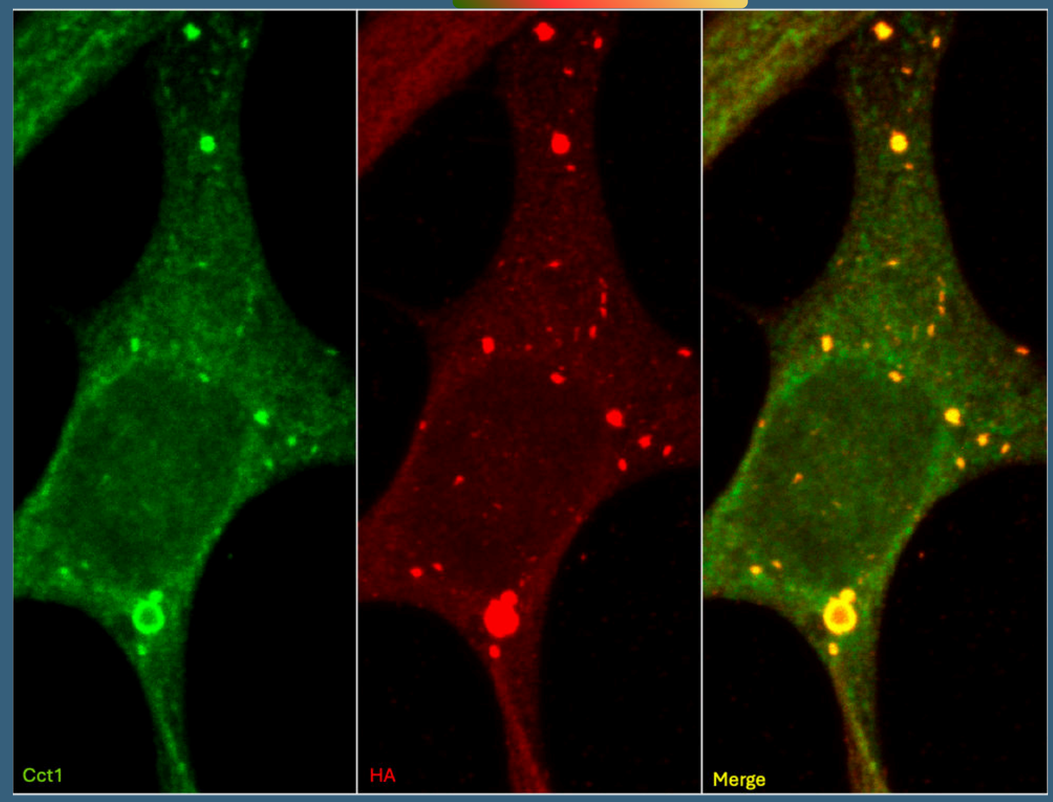
Impact of mutant Huntingtin

3D Z-STACKED FLUORESCENCE MICROSCOPY (.LIF FORMAT).

Channels used:

- Green = CCT1
- Red = mHTT-HA
- Blue = Nucleus (optional)

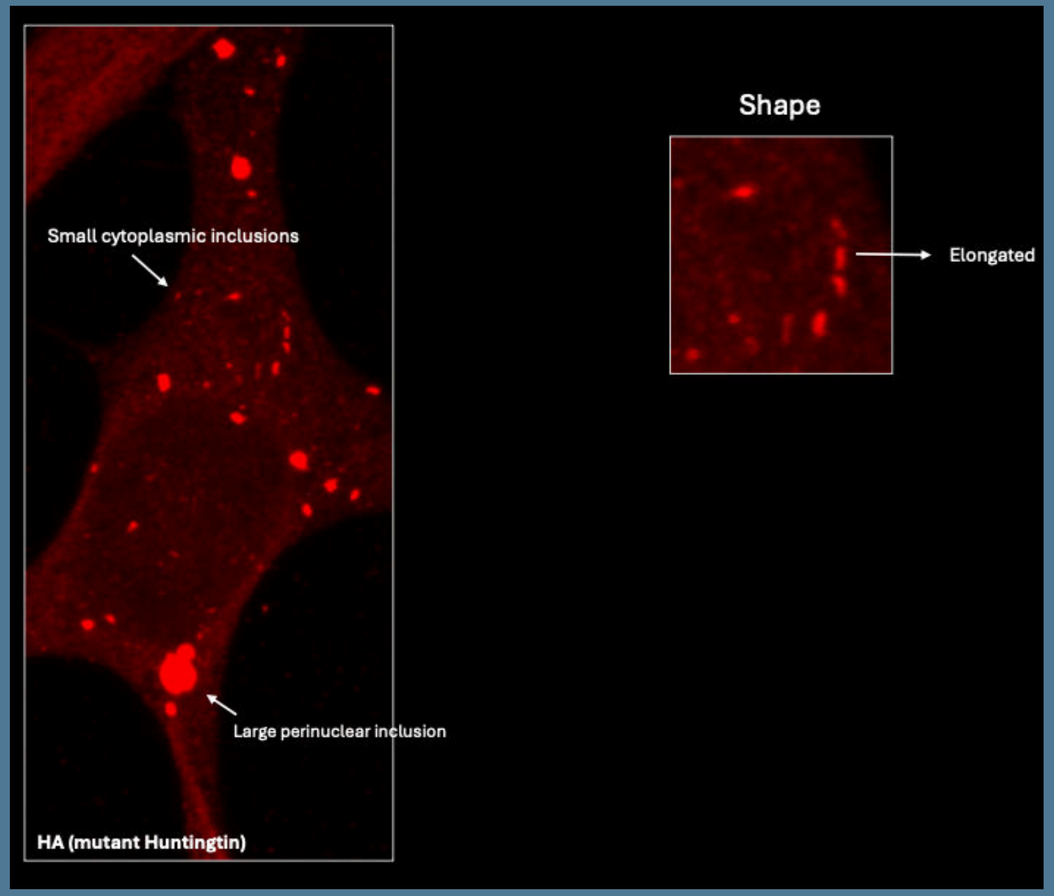
~100 annotated images.



Annotated example

annotated image hints at:

- Big mHTT aggregate near the nucleus, potenetially surrounded by CCT1 signal
- Smaller puncta in the cytoplasm.
- Elongated mHTT aggregates



Overview of the pipeline

🧬 COLOCALIZATION ANALYSIS PIPELINE

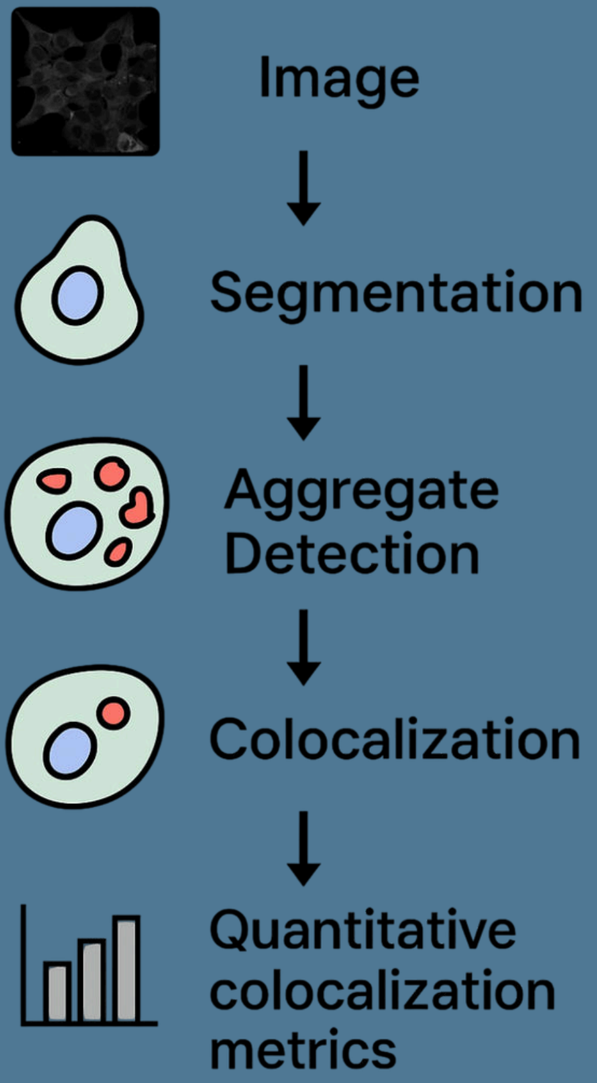
- MODULAR STEPS:
  A. IMAGE PREPROCESSING
    - SPLIT INTO Z-SLICES
    - NORMALIZATION.

  B. SEGMENTATION
    - THRESHOLDING MHTT
    - MORPHOLOGICAL OPERATIONS

  C. AGGREGATE DETECTION
    - ALIGNS MHTT AND CCT1 SEGMENTATIONS TO INDIVIDUAL CELLS AND NUCLEI (IF APPLICABLE).

  D. COLOCALIZATION ANALYSIS
    - INTENSITY-BASED METRICS: PEARSON'S, MANDERS' COEFFICIENTS.
    - SEGMENTATION-BASED METRICS: DICE COEFFICIENT, IOU.
    - STATISTICAL ENRICHMENT: MANN–WHITNEY U TEST, COHEN'S D.

  - DESIGN PRINCIPLES: HUMAN-IN-THE-LOOP, TRANSPARENCY, BIOLOGICAL INTERPRETABILITY.



Structural Colocalization analysis

What each metrics measures

Metric	Interpretation
Dice & IoU	Physical overlap between segmented mHTT and CCT1 regions.
Manders’ M1 & M2	Directional overlap – how much one protein’s signal overlaps with the other.
Pearson Correlation	How similarly the intensity values vary across the image space.

Structural Colocalization analysis

Interpretation of results

Finding	What It Tells Us
Strong overall overlap (Dice, IoU, Manders)	CCT1 is consistently found at the location of mHTT aggregates.
Asymmetric colocalization (Manders M1 low, M2 high)	Most CCT1 overlaps with mHTT, but not all mHTT aggregates contain CCT1.
Selective recruitment	Suggests CCT1 is targeting specific aggregates.

Enrichment analysis

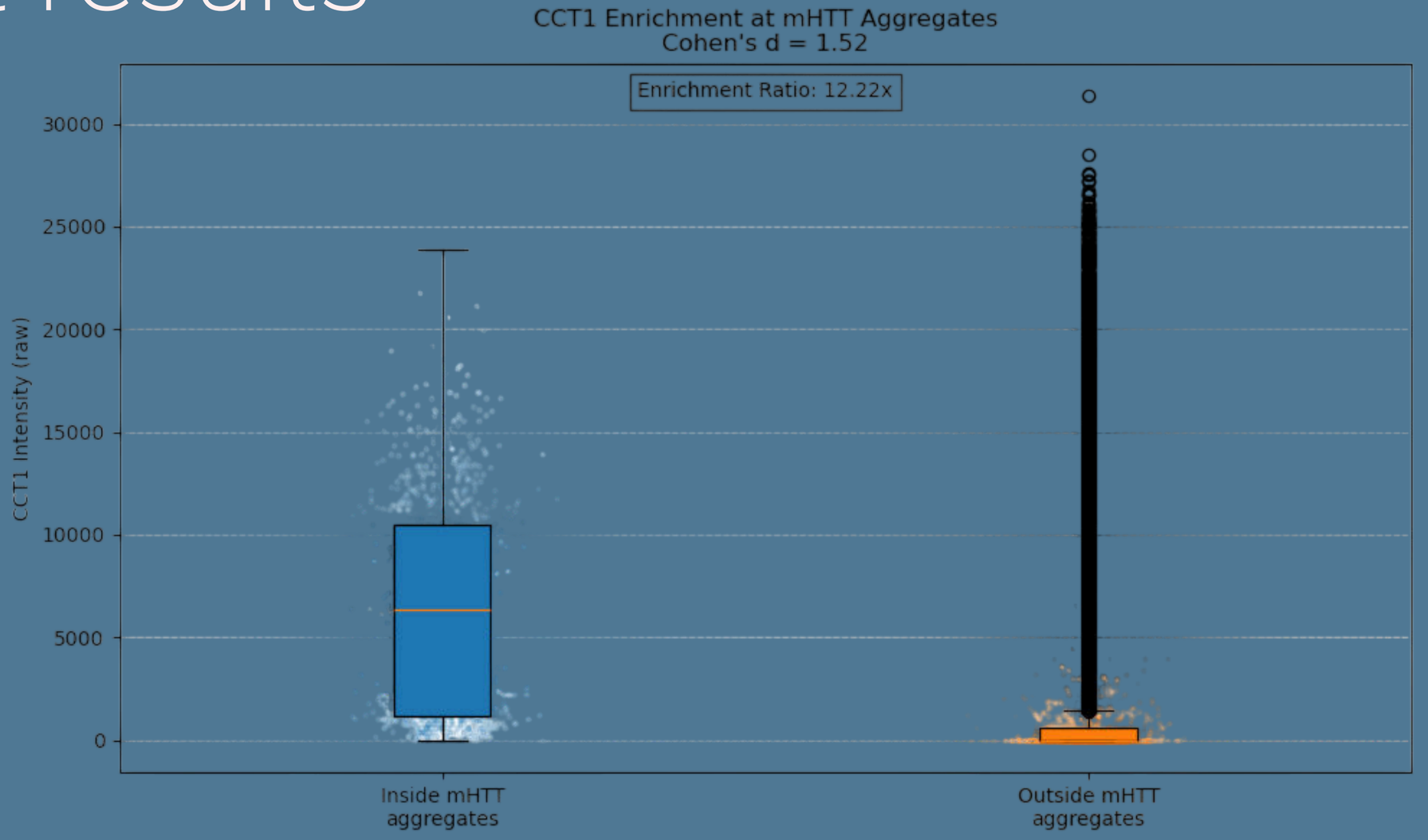
What each metrics measures

Metric	Interpretation
Mann-Whitney U test	Is CCT1 higher inside aggregates?
Cohen’s d	How strong is the difference?
Fold change	How many times higher inside?

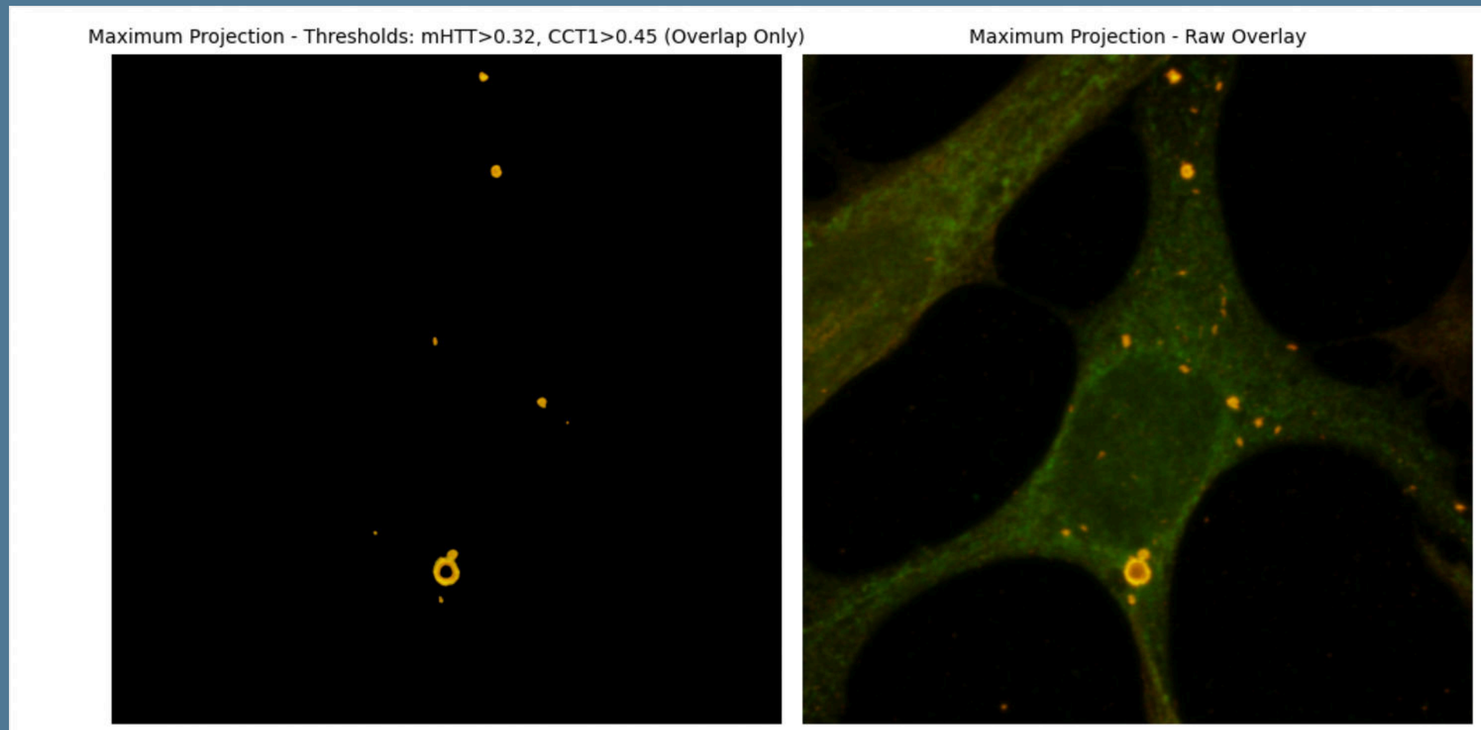
Intensity-based enrichment results

Mann–Whitney U test (3.6×10^{6}) shows significant enrichment.
Cohen's d > 1.52: large effect size.
Result: ~12× higher CCT1 signal inside aggregates.

- Otsu thresholding used to segment mHTT aggregates.
- CCT1 intensity compared inside vs. outside mHTT regions.



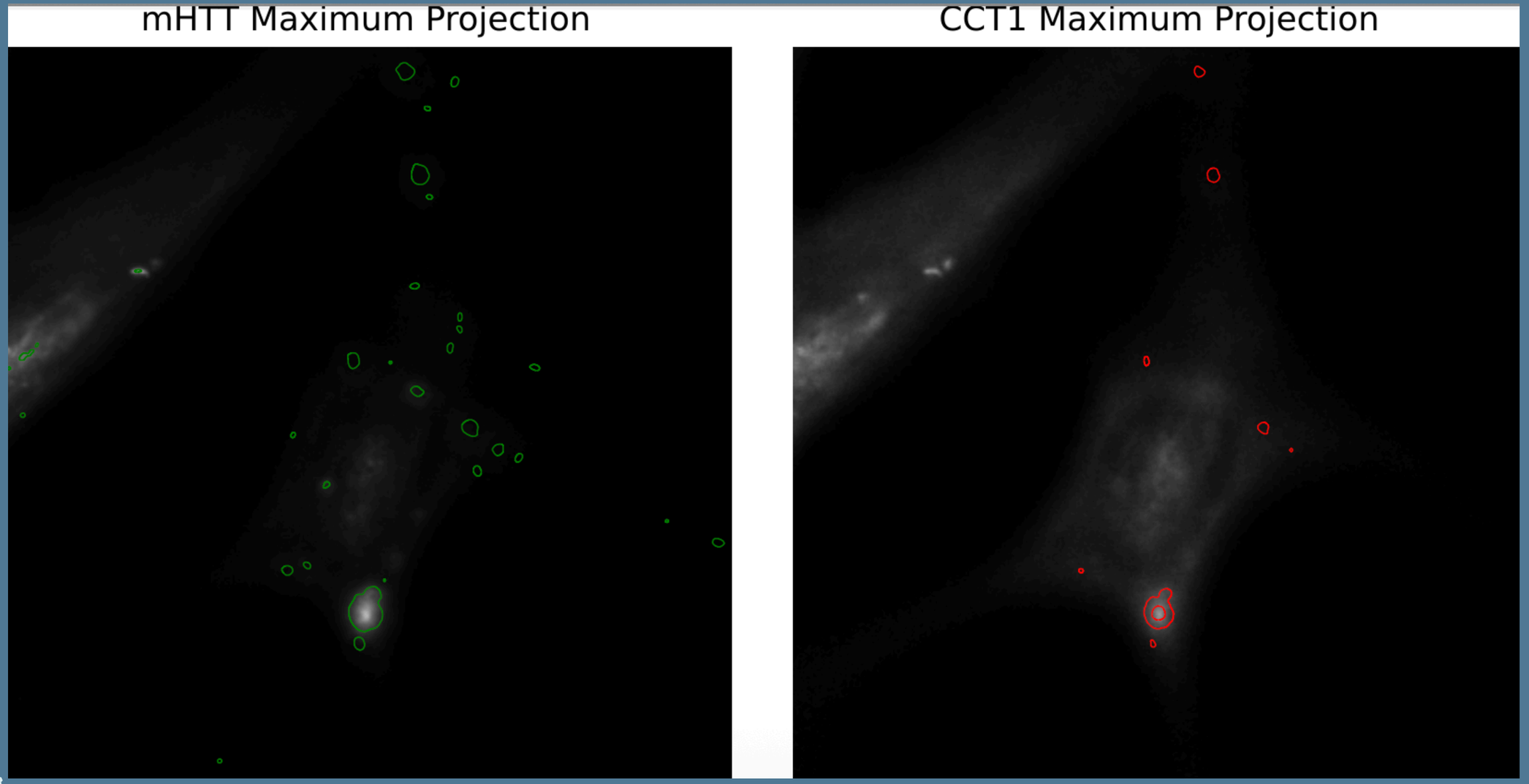
Visual confirmation




```
mHTT area: 23407 pixels
CCT1 area: 8367 pixels
Overlap area: 8367 pixels
Overlap percentage: 35.75% of mHTT, 100.00% of CCT1
Colocalized aggregates: 8

===== COLOCALIZATION ANALYSIS RESULTS =====
Parameters: mHTT factor=4.0, CCT1 factor=4.5, sigma=3.5, min_size=20
Thresholds: mHTT>0.3212, CCT1>0.4521
1. Pearson's Correlation: r = 0.8454 (p=0.00e+00)
2. Manders' Coefficients: M1=0.4779, M2=1.0000
3. Dice Similarity: 0.5267
4. Mann-Whitney U: U=3603453.00, p=0.00e+00
5. Cohen's d: d=2.6439, mean_in=0.81, mean_out=0.49
6. Intersection over Union (IoU): 0.3575
```

Visual confirmation

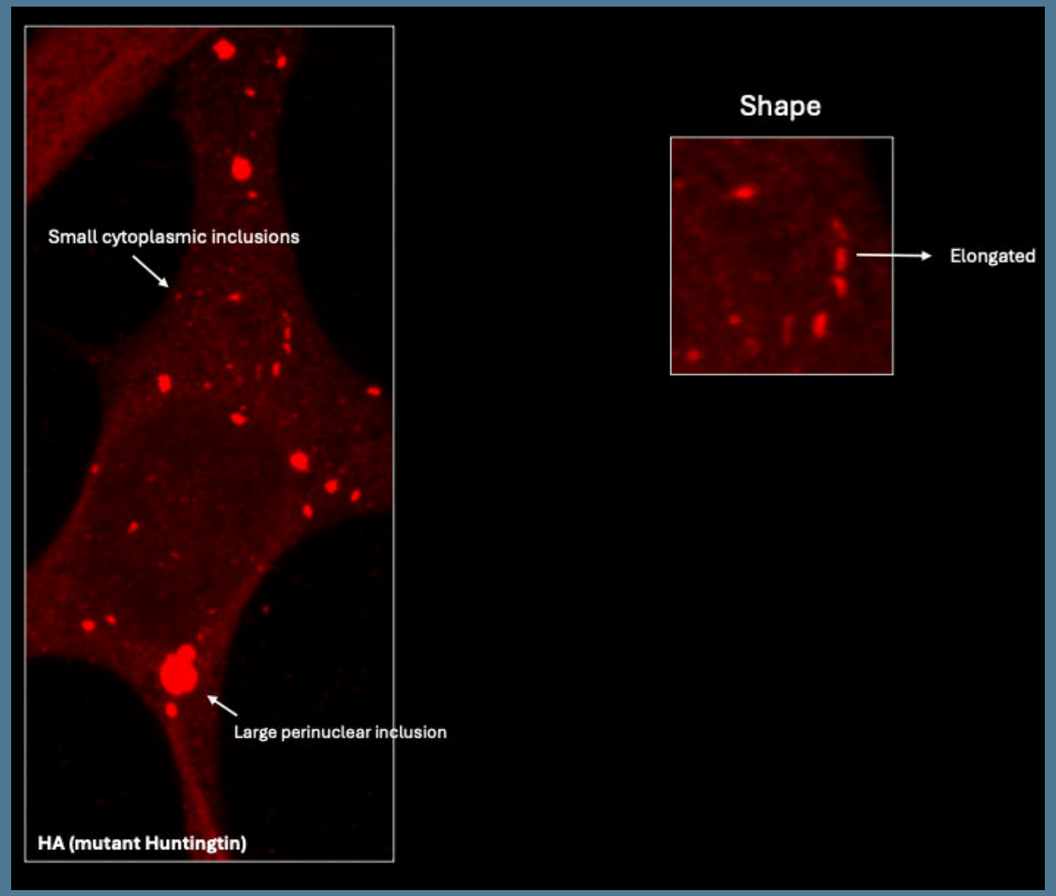


Visual confirmation

Visualizations show:

- CCT1 wrapping around large mHTT inclusions.
- Smaller puncta with full signal overlap.

Confirms patterns seen in quantitative metrics.

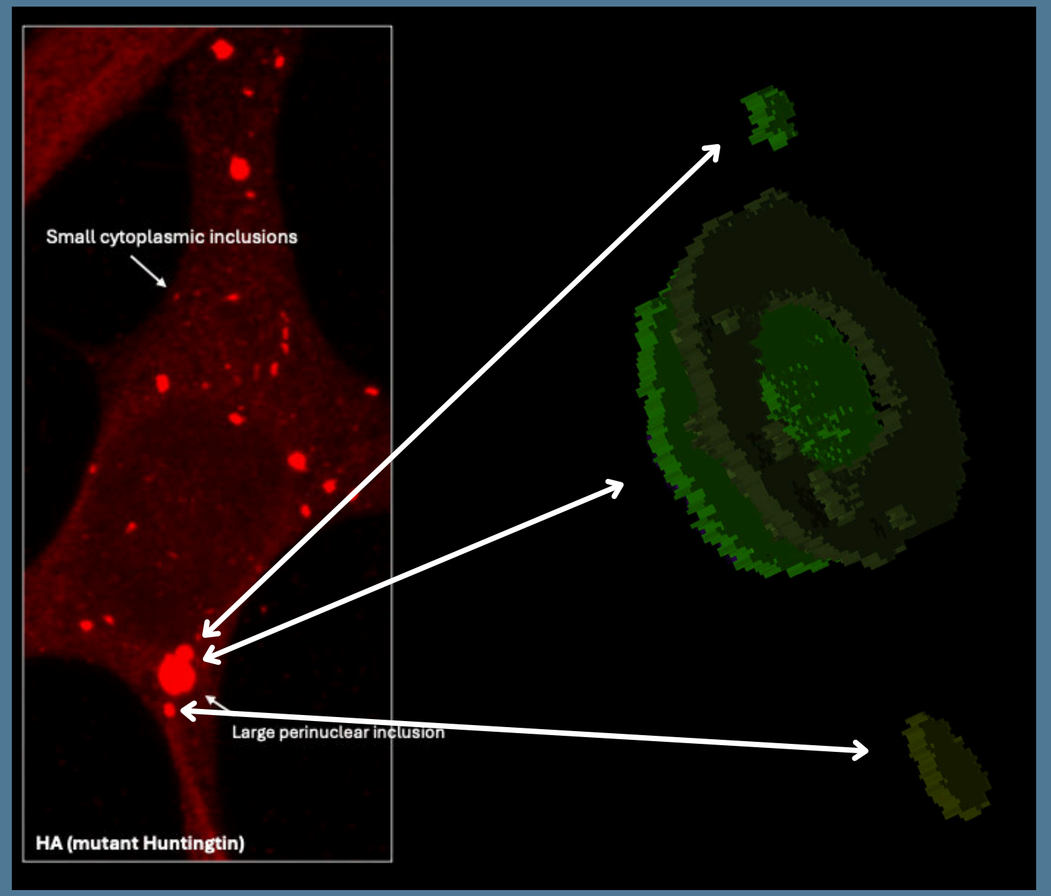


Visual confirmation

3D visualization of CCT1 areas shows:

- CCT1 wrapping around large mHTT inclusions.
- Smaller puncta with full signal overlap.

Confirms patterns seen in quantitative metrics.



Conclusion

CCT1 is consistently enriched at mHTT aggregation sites.

This holds true across:

- Intensity-based enrichment and structural overlap metrics
- Both pixel-wise statistics and object-level spatial analyses

all on

Asymmetric colocalization indicates selective recruitment of CCT1 to mHTT aggregates

The developed pipeline is:

- Modular, automated, and biologically interpretable
- Designed for non-programmers via a user-friendly GUI
- Ready for scalable, high-throughput 3D colocalization research
- Includes safeguards for interpretability and human-in-the-loop oversight

Discussion

Ethical & Practical Implications:

✅ Reproducibility
Automation reduces user bias, fatigue, and increases consistency.

✅ Transparency
Version logging, overlay inspection, and hybrid AI + human review maintains interpretability.

⚠ Over-reliance Risk
High overlap scores may give false confidence → visual overlays remain mandatory.

⚠ Data Bias
Segmentation models may fail on new data → re-validation required.

⚠ Cross-disciplinary Accountability
Ongoing dialogue between data scientists and biologists is essential to detect implausible results.

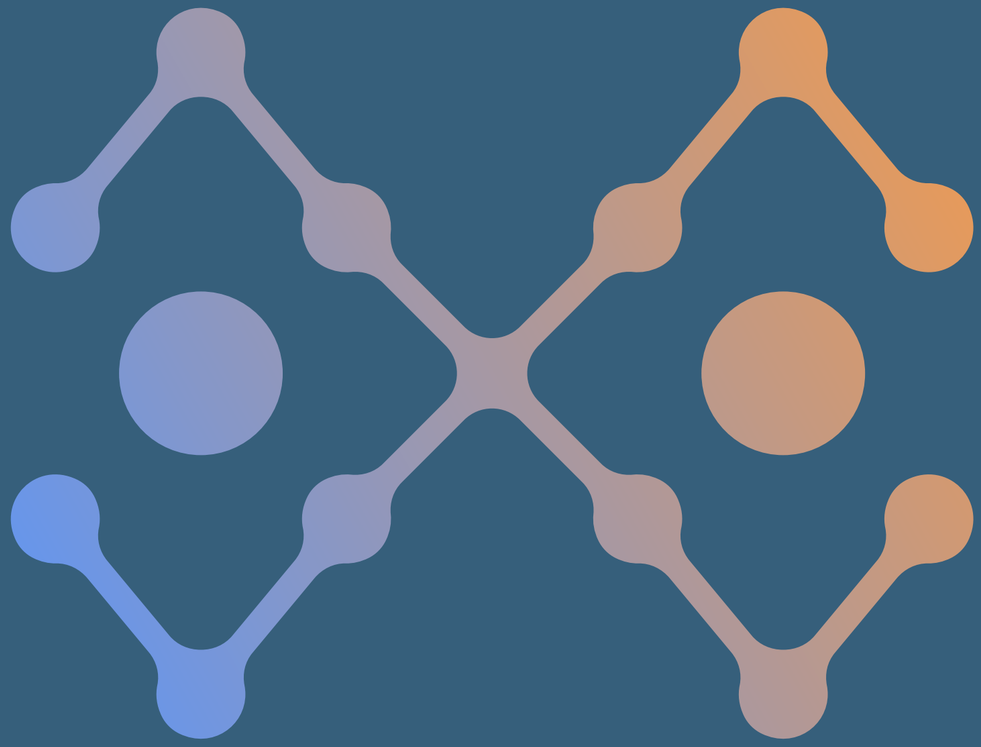
Discussion

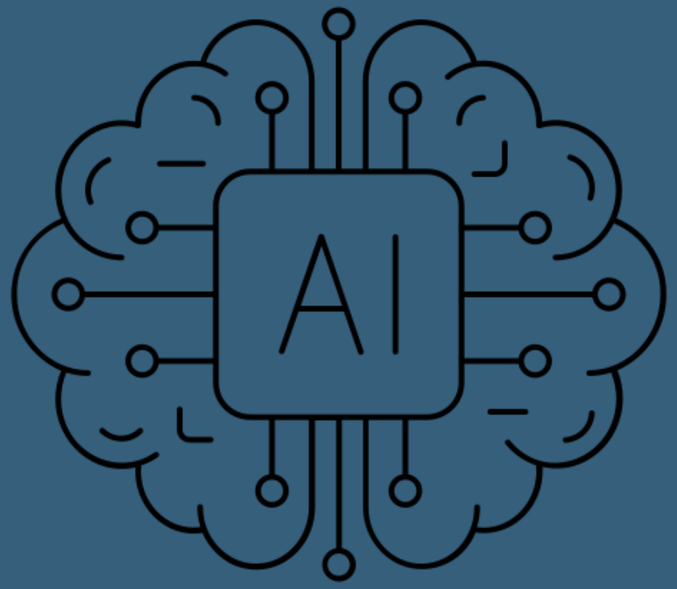
Future Extensions of the Pipeline:

- 🩹 Larger-scale biological validation to confirm enrichment generalizability.
- 🧬 Improved segmentation models with expanded training data.
- ⚠️ Automated image quality monitoring to detect data deviations (using metadata).
- 🫓 Per-cell and nuclear segmentation for finer resolution.
- ⚪ Oligomeric species analysis using additional A11 channel data.

The END







learned segmentation

- nnU-Net trained on mHTT and CCT1 masks.
- Enrichment decreased to ~1.83×
- (Cohen’s d = 0.42).
- inadequate

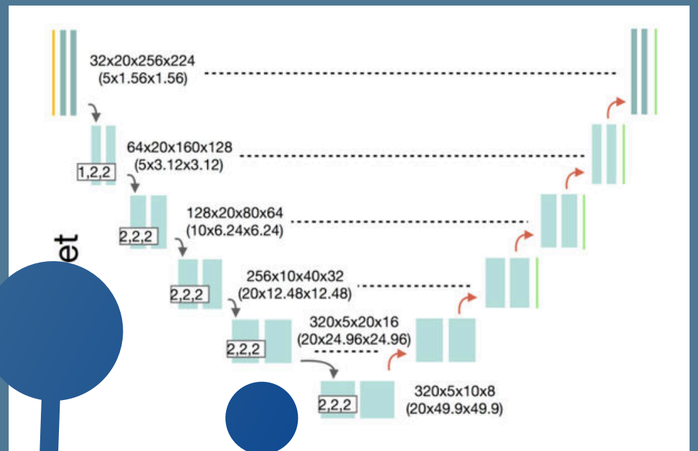


Figure: nnU-Net segmentation highlights colocalization zones. Displaying broader aggregate boundaries lower enrichment magnitude

